Relative Efficiency. If W_1 and W_2 are two unbiased estimators of θ, W_1 is efficient relative to W_2 when $\text{Var}(W_1) \leq \text{Var}(W_2)$ for all θ, with strict inequality for at least one value of θ.

Earlier, we showed that, for estimating the population mean μ, $\text{Var}(\bar{Y}) < \text{Var}(Y_1)$ for any value of σ^2 whenever $n > 1$. Thus, $\bar{Y}$ is efficient relative to Y_1 for estimating μ. We cannot always choose between unbiased estimators based on the smallest variance criterion: given two unbiased estimators of θ, one can have smaller variance from some values of θ, while the other can have smaller variance for other values of θ.

If we restrict our attention to a certain class of estimators, we can show that the sample average has the smallest variance. Problem C.2 asks you to show that $\bar{Y}$ has the smallest variance among all unbiased estimators that are also linear functions of $Y_1, Y_2, \ldots, Y_n$. The assumptions are that the Y_i have common mean and variance, and that they are pairwise uncorrelated.

If we do not restrict our attention to unbiased estimators, then comparing variances is meaningless. For example, when estimating the population mean μ, we can use a trivial estimator that is equal to zero, regardless of the sample that we draw. Naturally, the variance of this estimator is zero (since it is the same value for every random sample). But the bias of this estimator is $-\mu$, so it is a very poor estimator when $|\mu|$ is large.

One way to compare estimators that are not necessarily unbiased is to compute the **mean squared error** (**MSE**) of the estimators. If W is an estimator of θ, then the MSE of W is defined as $\text{MSE}(W) = \text{E}[(W - \theta)^2]$. The MSE measures how far, on average, the estimator is away from θ. It can be shown that $\text{MSE}(W) = \text{Var}(W) + [\text{Bias}(W)]^2$, so that $\text{MSE}(W)$ depends on the variance and bias (if any is present). This allows us to compare two estimators when one or both are biased.

C.3 Asymptotic or Larger Sample Properties of Estimators

In Section C.2, we encountered the estimator Y_1 for the population mean μ, and we saw that, even though it is unbiased, it is a poor estimator because its variance can be much larger than that of the sample mean. One notable feature of Y_1 is that it has the same variance for any sample size. It seems reasonable to require any estimation procedure to improve as the sample size increases. For estimating a population mean μ, $\bar{Y}$ improves in the sense that its variance gets smaller as n gets larger; Y_1 does not improve in this sense.

We can rule out certain silly estimators by studying the *asymptotic* or *large sample* properties of estimators. In addition, we can say something positive about estimators that are not unbiased and whose variances are not easily found.

Asymptotic analysis involves approximating the features of the sampling distribution of an estimator. These approximations depend on the size of the sample. Unfortunately, we are necessarily limited in what we can say about how "large" a sample size is needed for asymptotic analysis to be appropriate; this depends on the underlying population distribution. But large sample approximations have been known to work well for sample sizes as small as $n = 20$.

Consistency

The first asymptotic property of estimators concerns how far the estimator is likely to be from the parameter it is supposed to be estimating as we let the sample size increase indefinitely.

Consistency. Let W_n be an estimator of θ based on a sample $Y_1, Y_2, \ldots, Y_n$ of size n. Then, W_n is a **consistent estimator** of θ if for every $\varepsilon > 0$,

$$P(|W_n - \theta| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty. \tag{C.7}$$

If W_n is not consistent for θ, then we say it is **inconsistent**.

When W_n is consistent, we also say that θ is the **probability limit** of W_n, written as $\text{plim}(W_n) = \theta$.

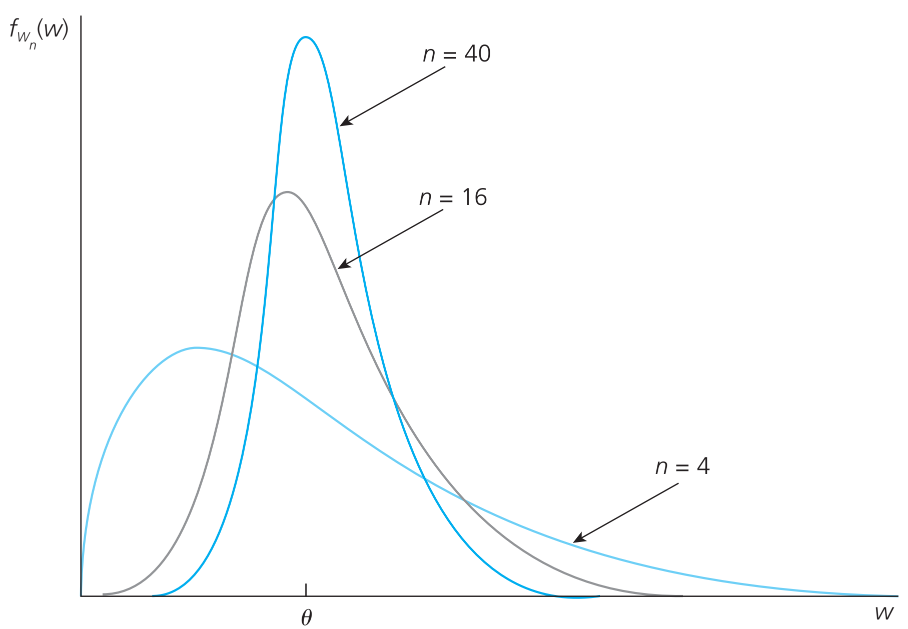
Unlike unbiasedness—which is a feature of an estimator for a given sample size—consistency involves the behavior of the sampling distribution of the estimator as the sample size n gets large. To emphasize this, we have indexed the estimator by the sample size in stating this definition, and we will continue with this convention throughout this section.

Equation (C.7) looks technical, and it can be rather difficult to establish based on fundamental probability principles. By contrast, interpreting (C.7) is straightforward. It means that the distribution of W_n becomes more and more concentrated about θ, which roughly means that for larger sample sizes, W_n is less and less likely to be very far from θ. This tendency is illustrated in Figure C.3.

If an estimator is not consistent, then it does not help us to learn about θ, even with an unlimited amount of data. For this reason, consistency is a minimal requirement of an estimator used in statistics or econometrics. We will encounter estimators that are consistent under certain assumptions and inconsistent when those assumptions fail. When estimators

FIGURE C.3

The sampling distributions of a consistent estimator for three sample sizes.



are inconsistent, we can usually find their probability limits, and it will be important to know how far these probability limits are from θ.

As we noted earlier, unbiased estimators are not necessarily consistent, but those whose variances shrink to zero as the sample size grows *are* consistent. This can be stated formally: If W_n is an unbiased estimator of θ and $\text{Var}(W_n) \rightarrow 0$ as $n \rightarrow \infty$, then $\text{plim}(W_n) = \theta$. Unbiased estimators that use the entire data sample will usually have a variance that shrinks to zero as the sample size grows, thereby being consistent.

A good example of a consistent estimator is the average of a random sample drawn from a population with μ and variance σ^2. We have already shown that the sample average is unbiased for μ. In equation (C.6), we derived $\text{Var}(\bar{Y}_n) = \sigma^2/n$ for any sample size n. Therefore, $\text{Var}(\bar{Y}_n) \rightarrow 0$ as $n \rightarrow \infty$, so $\bar{Y}_n$ is a consistent estimator of μ (in addition to being unbiased).

The conclusion that $\bar{Y}_n$ is consistent for μ holds even if $\text{Var}(\bar{Y}_n)$ does not exist. This classic result is known as the **law of large numbers** (**LLN**).

Law of Large Numbers. Let $Y_1, Y_2, \ldots, Y_n$ be independent, identically distributed random variables with mean μ. Then,

$$\text{plim}(\bar{Y}_n) = \mu. \tag{C.8}$$

The law of large numbers means that, if we are interested in estimating the population average μ, we can get arbitrarily close to μ by choosing a sufficiently large sample. This fundamental result can be combined with basic properties of plims to show that fairly complicated estimators are consistent.

Property PLIM.1: Let θ be a parameter and define a new parameter, $\gamma = g(\theta)$, for some *continuous* function $g(\theta)$. Suppose that $\text{plim}(W_n) = \theta$. Define an estimator of γ by $G_n = g(W_n)$. Then,

$$\text{plim}(G_n) = \gamma. \tag{C.9}$$

This is often stated as

$$\text{plim } g(W_n) = g(\text{plim } W_n) \tag{C.10}$$

for a continuous function $g(\theta)$.

The assumption that $g(\theta)$ is continuous is a technical requirement that has often been described nontechnically as "a function that can be graphed without lifting your pencil from the paper." Because all the functions we encounter in this text are continuous, we do not provide a formal definition of a continuous function. Examples of continuous functions are $g(\theta) = a + b\theta$ for constants a and b, $g(\theta) = \theta^2$, $g(\theta) = 1/\theta$, $g(\theta) = \sqrt{\theta}$, $g(\theta) = \exp(\theta)$, and many variants on these. We will not need to mention the continuity assumption again.

As an important example of a consistent but biased estimator, consider estimating the standard deviation, σ, from a population with mean μ and variance σ^2. We already claimed that the sample variance $S_n^2 = (n-1)^{-1}\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2$ is unbiased for σ^2. Using the law of large numbers and some algebra, S_n^2 can also be shown to be consistent for σ^2.

The natural estimator of $\sigma = \sqrt{\sigma^2}$ is $S_n = \sqrt{S_n^2}$ (where the square root is always the positive square root). S_n, which is called the **sample standard deviation**, is *not* an unbiased estimator because the expected value of the square root is *not* the square root of the expected value (see Section B.3). Nevertheless, by PLIM.1, plim $S_n = \sqrt{\text{plim } S_n^2} = \sqrt{\sigma^2} = \sigma$, so S_n is a consistent estimator of σ.

Here are some other useful properties of the probability limit:

Property PLIM.2: If $\text{plim}(T_n) = \alpha$ and $\text{plim}(U_n) = \beta$, then

(i) $\text{plim}(T_n + U_n) = \alpha + \beta$;
(ii) $\text{plim}(T_n U_n) = \alpha\beta$;
(iii) $\text{plim}(T_n/U_n) = \alpha/\beta$, provided $\beta \neq 0$.

These three facts about probability limits allow us to combine consistent estimators in a variety of ways to get other consistent estimators. For example, let $\{Y_1, \ldots, Y_n\}$ be a random sample of size n on annual earnings from the population of workers with a high school education and denote the population mean by μ_Y. Let $\{Z_1, \ldots, Z_n\}$ be a random sample on annual earnings from the population of workers with a college education and denote the population mean by μ_Z. We wish to estimate the percentage difference in annual earnings between the two groups, which is $\gamma = 100\cdot(\mu_Z - \mu_Y)/\mu_Y$. (This is the percentage by which average earnings for college graduates differs from average earnings for high school graduates.) Because $\bar{Y}_n$ is consistent for μ_Y and $\bar{Z}_n$ is consistent for μ_Z, it follows from PLIM.1 and part (iii) of PLIM.2 that

$$G_n \equiv 100 \cdot (\bar{Z}_n - \bar{Y}_n)/\bar{Y}_n$$

is a consistent estimator of γ. G_n is just the percentage difference between $\bar{Z}_n$ and $\bar{Y}_n$ in the sample, so it is a natural estimator. G_n is not an unbiased estimator of γ, but it is still a good estimator except possibly when n is small.

Asymptotic Normality

Consistency is a property of point estimators. Although it does tell us that the distribution of the estimator is collapsing around the parameter as the sample size gets large, it tells us essentially nothing about the *shape* of that distribution for a given sample size. For constructing interval estimators and testing hypotheses, we need a way to approximate the distribution of our estimators. Most econometric estimators have distributions that are well approximated by a normal distribution for large samples, which motivates the following definition.

Asymptotic Normality. Let $\{Z_n: n = 1, 2, \ldots\}$ be a sequence of random variables, such that for all numbers z,

$$P(Z_n \leq z) \rightarrow \Phi(z) \text{ as } n \rightarrow \infty, \qquad \text{C.11}$$

where $\Phi(z)$ is the standard normal cumulative distribution function. Then, Z_n is said to have an *asymptotic standard normal distribution*. In this case, we often write $Z_n \overset{a}{\sim} \text{Normal}(0,1)$. (The "*a*" above the tilde stands for "asymptotically" or "approximately.")

Property (C.11) means that the cumulative distribution function for Z_n gets closer and closer to the cdf of the standard normal distribution as the sample size n gets large. When

asymptotic normality holds, for large n we have the approximation $P(Z_n \leq z) \approx \Phi(z)$. Thus, probabilities concerning Z_n can be approximated by standard normal probabilities.

The **central limit theorem** (**CLT**) is one of the most powerful results in probability and statistics. It states that the average from a random sample for *any* population (with finite variance), when standardized, has an asymptotic standard normal distribution.

Central Limit Theorem. Let $\{Y_1, Y_2, \ldots, Y_n\}$ be a random sample with mean μ and variance σ^2. Then,

$$Z_n = \frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \tag{C.12}$$

has an asymptotic standard normal distribution.

The variable Z_n in (C.12) is the standardized version of $\bar{Y}_n$: we have subtracted off $E(\bar{Y}_n) = \mu$ and divided by $sd(\bar{Y}_n) = \sigma/\sqrt{n}$. Thus, regardless of the population distribution of Y, Z_n has mean zero and variance one, which coincides with the mean and variance of the standard normal distribution. Remarkably, the entire distribution of Z_n gets arbitrarily close to the standard normal distribution as n gets large.

We can write the standardized variable in equation (C.12) as $\sqrt{n}(\bar{Y}_n - \mu)/\sigma$, which shows that we must multiply the difference between the sample mean and the population mean by the square root of the sample size in order to obtain a useful limiting distribution. Without the multiplication by $\sqrt{n}$, we would just have $(\bar{Y}_n - \mu)/\sigma$, which converges in probability to zero. In other words, the distribution of $(\bar{Y}_n - \mu)/\sigma$ simply collapses to a single point as $n \to \infty$, which we know cannot be a good approximation to the distribution of $(\bar{Y}_n - \mu)/\sigma$ for reasonable sample sizes. Multiplying by $\sqrt{n}$ ensures that the variance of Z_n remains constant. Practically, we often treat $\bar{Y}_n$ as being approximately normally distributed with mean μ and variance σ^2/n, and this gives us the correct statistical procedures because it leads to the standardized variable in equation (C.12).

Most estimators encountered in statistics and econometrics can be written as functions of sample averages, in which case we can apply the law of large numbers and the central limit theorem. When two consistent estimators have asymptotic normal distributions, we choose the estimator with the smallest asymptotic variance.

In addition to the standardized sample average in (C.12), many other statistics that depend on sample averages turn out to be asymptotically normal. An important one is obtained by replacing σ with its consistent estimator S_n in equation (C.12):

$$\frac{\bar{Y}_n - \mu}{S_n/\sqrt{n}} \tag{C.13}$$

also has an approximate standard normal distribution for large n. The exact (finite sample) distributions of (C.12) and (C.13) are definitely not the same, but the difference is often small enough to be ignored for large n.

Throughout this section, each estimator has been subscripted by n to emphasize the nature of asymptotic or large sample analysis. Continuing this convention clutters the notation without providing additional insight, once the fundamentals of asymptotic analysis are understood. Henceforth, we drop the n subscript and rely on you to remember that estimators depend on the sample size, and properties such as consistency and asymptotic normality refer to the growth of the sample size without bound.